**Project Design Phase-II**
**Technology Stack (Architecture & Stack)**

| | |
|---|---|
| Date | 23 MARCH 2026 |
| Team ID | 69c0d503f58a227701cdee5a |
| Project Name | AUTOMATING EMPLOYEE DATA |
| Maximum Marks | 4 Marks |

**Technical Architecture (Explanation)**

**Architecture Type:**

**3-Tier Architecture (Client–Server Model)**

**Flow:**

1. **User interacts with Web UI (Frontend)**
2. **Request goes to Application Server (Backend Logic)**
3. **Backend processes data and interacts with Database**
4. **Response is sent back to user**

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**Simple Architecture Diagram (You can draw this)**

**[ User (Browser)]**
↓
**[ Frontend (HTML, CSS, JS)]**
↓
**[ Backend Server (PHP / Node.js)]**
↓
**[ Database (MySQL)]**
↓

Guidelines:

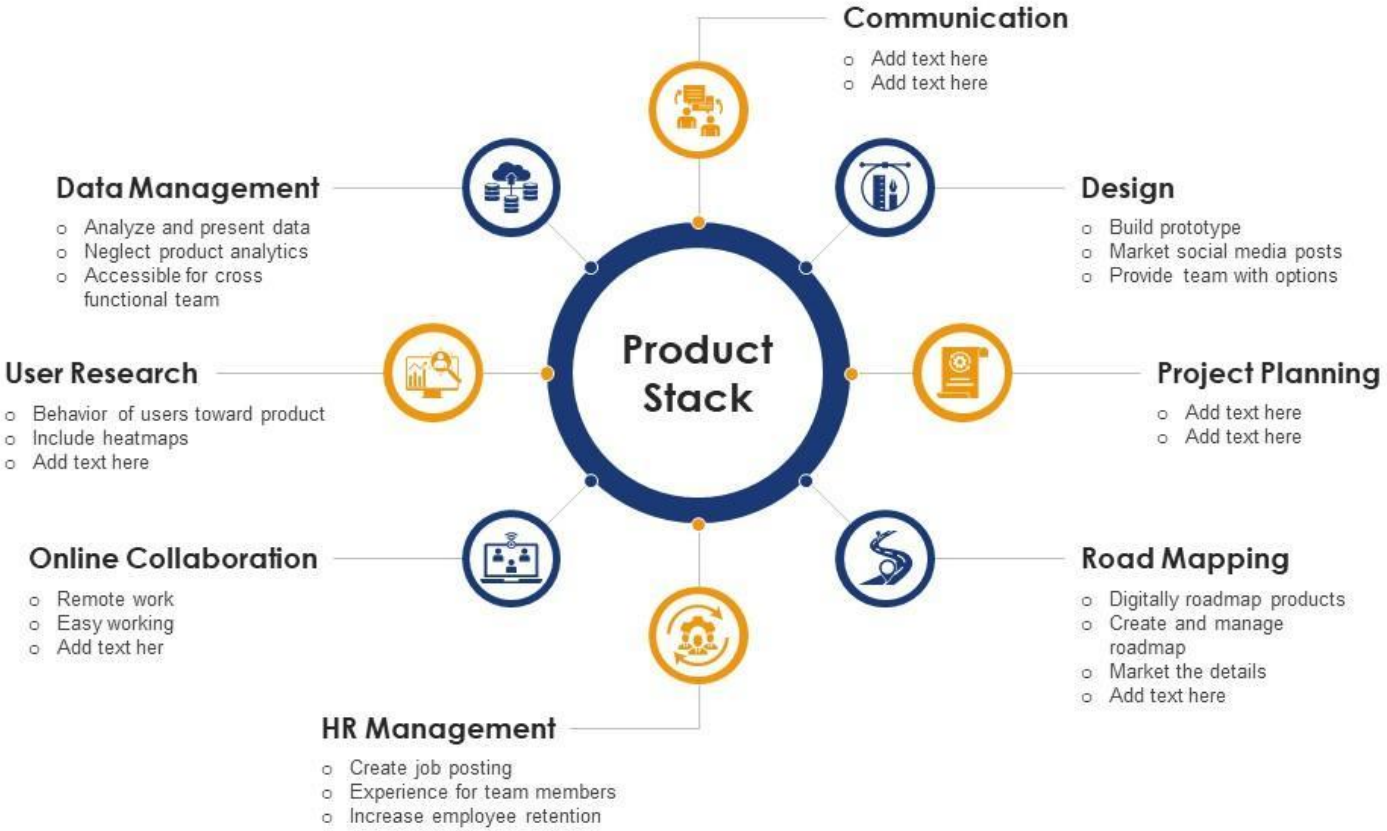
- Store employee details in system
- Manage data (Add, Edit, Delete)

The system helps to manage employee data easily, reduces manual work, and improves efficiency.”

- User Registration
- Login System
- Add Employee
- Edit Employee
- Delete Employee
- View Employee List

# Areas of Project Product Tech Stack

This slide showcases the areas of product tech stack. Its aim is to improve these mentioned areas for a feasible and manageable tech stack to increase efficiency. This slide includes communication, design, project planning, road mapping, HR management, online collaboration, user research, etc.




This slide is 100% editable. Adapt it to your need and capture your audience's attention.

**Table-1 : Components & Technologies:**

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1. | User Interface | How user interacts with application e.g.<br>Web UI, Mobile App, Chatbot etc. | HTML, CSS, JavaScript / Angular Js / React Js etc. |
| 2. | Application Logic-1 | Logic for a process in the application | Java / Python |
| 3. | Application Logic-2 | Logic for a process in the application | IBM Watson STT service |
| 4. | Application Logic-3 | Logic for a process in the application | IBM Watson Assistant |
| 5. | Database | Data Type, Configurations etc. | MySQL, NoSQL, etc. |
| 6. | Cloud Database | Database Service on Cloud | IBM DB2, IBM Cloudant etc. |
| 7. | File Storage | File storage requirements | IBM Block Storage or Other Storage Service or Local Filesystem |
| 8. | External API-1 | Purpose of External API used in the application | IBM Weather API, etc. |
| 9. | External API-2 | Purpose of External API used in the application | Aadhar API, etc. |
| 10. | Machine Learning Model | Purpose of Machine Learning Model | Object Recognition Model, etc. |
| 11. | Infrastructure (Server / Cloud) | Application Deployment on Local System / Cloud<br>Local Server Configuration:<br>Cloud Server Configuration : | Local, Cloud Foundry, Kubernetes, etc. |

**Table-2: Application Characteristics:**

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 1. | Open-Source Frameworks | List the open-source frameworks used | Technology of Opensource framework |
| 2. | Security Implementations | List all the security / access controls implemented, use of firewalls etc. | e.g. SHA-256, Encryptions, IAM Controls, OWASP etc. |
| 3. | Scalable Architecture | Justify the scalability of architecture (3 – tier, Micro-services) | Technology used |
| 4. | Availability | Justify the availability of application (e.g. use of load balancers, distributed servers etc.) | Technology used |
| 5. | Performance | Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc. | Technology used |